

KEYSTRA ENERGY GROUP

SOLUTION 2: PIPELINE INTEGRITY AND OPERATIONAL RISK INTELLIGENCE

Portfolio Project | Proof of Concept

Data Source: PHMSA Hazardous Liquid Accident Data, U.S. Department of Transportation — public U.S. regulatory dataset

PROJECT OVERVIEW

Pipeline Integrity and Operational Risk Intelligence is a portfolio proof-of-concept solution developed for Keystra Energy Group, a fictional integrated energy company. The project demonstrates how data analytics can be used to identify pipeline risk patterns, assess their potential impact, and translate insights into practical integrity priorities.

The solution examines incident trends, geographic and asset exposure, failure mechanisms, and financial, environmental, and safety consequences, culminating in two dashboards: The Executive Pipeline Risk Overview for senior management, and the Pipeline Integrity and Operational Intelligence dashboard for integrity engineers and maintenance planners.

The project is not presented as a solution deployed or tested in a real organisation. Instead, it demonstrates how analytics can support a more proactive, evidence-based approach to pipeline integrity management.

BUSINESS CHALLENGE

Pipeline networks contain assets that vary in age, design, operating conditions, location, and exposure, while resources for inspection, maintenance, repairs, and investment are limited. The key challenge is therefore determining where risk is greatest, what is driving it, and where intervention should be prioritised.

Simply recording incidents does not provide enough insight. Organisations need to understand recurring patterns across locations, assets, failure mechanisms, and consequences to distinguish frequent operational risks from less frequent events capable of causing significant financial, environmental, or safety impacts. This analysis addresses the challenge by transforming historical incident information into structured risk intelligence that can support targeted inspection, maintenance prioritisation, preventive interventions, and stronger pipeline integrity management.

Dataset and Data Preparation

This analysis was done using the PHMSA Hazardous Liquid Accident Data, a real world public U.S. regulatory dataset published by the Pipeline and Hazardous Materials Safety Administration (PHMSA), an agency of the U.S. Department of Transportation. The dataset contains hazardous liquid pipeline incidents reported by operators across the United

States, with the version used covering January 2010 through 2026 and providing more than fifteen years of historical incident data.

The dataset was selected because it captures a broad range of information including incident timing and location, operators, facility types, pipeline characteristics, failure causes, product releases, financial losses, environmental impacts, and safety outcomes. This provided the foundation for analysing incident trends, geographic and asset exposure, failure mechanisms, and their wider consequences. Because the dataset was originally designed for regulatory reporting rather than business analytics, it required preparation before analysis. Using Microsoft Excel and Power Query, the initial file was profiled at 5,959 incidents across 648 variables. Relevant fields were retained, data types and missing values were reviewed, and duplicate incident identifiers were checked. Regulatory and administrative fields that did not support the project objectives were removed, reducing the dataset to 38 analytical variables.

The refined dataset was subsequently exported to MySQL for database engineering and SQL analysis, and later used to develop the project's two pipeline risk dashboards. The PHMSA data remain public U.S. source data and were not generated by Keystra; the analysis is presented strictly as a portfolio demonstration and proof-of-concept, not as a solution tested or deployed within a real company.

ANALYTICAL APPROACH

The analysis was structured into six investigations, each building on the previous one to develop a broader understanding of pipeline risk. The investigations examine incident patterns, geographic exposure, asset characteristics, failure mechanisms, consequences, and strategic integrity priorities. Together, they demonstrate how historical operational data can be transformed into actionable pipeline risk intelligence.

Investigation 1: Understanding the Incident Landscape

The purpose of this investigation was to first understand the overall pipeline incident environment before looking into the specific causes and consequences of failures. Examining the volume, trend, and distribution of incidents provided a baseline for understanding how frequently failures occur, how the risk profile has changed over time, and where incidents appear to be more concentrated. The analysis showed that pipeline incidents remain a recurring operational challenge, despite the decline in reported incidents after the 2015 peak. It also showed that incidents are not evenly distributed across the industry. Certain areas accounted for a much larger share of reported incidents, reinforcing the importance of considering pipeline infrastructure and operational exposure when assessing risk.

Another important takeaway was the role of reliable incident data. Consistent records make it possible to identify recurring patterns, monitor changes over time, and move beyond reacting to individual incidents towards understanding broader risk trends.

Relevance to Nigeria

This provides a useful lesson for pipeline operations in Nigeria. Before deciding where to focus integrity resources, there is value in first establishing a clear picture of the local incident landscape. Analysing historical incidents alongside pipeline location, asset characteristics, operating activity, and other exposure factors could help identify areas requiring greater attention.

The U.S. experience therefore demonstrates the value of using data to understand where pipeline risk exists before deciding how to manage it. The specific patterns observed in the United States should not be assumed to apply directly to Nigeria; instead, the same analytical approach can be applied to Nigerian data to support more targeted inspection, maintenance, monitoring, and integrity investment.

Investigation 2: Geographic Risk Assessment

The purpose of this investigation was to understand how pipeline risk varies geographically and whether areas with the most incidents are necessarily the areas facing the greatest overall risk. Rather than relying on incident frequency alone, the analysis considered environmental, financial, and safety consequences to provide a broader view of geographic exposure.

The findings showed that different regions can have very different risk profiles. Texas had the highest incident frequency and environmental exposure, making it a clear operational hotspot. However, states such as West Virginia and Michigan showed that a lower number of incidents can still result in significant environmental or financial consequences. This demonstrates why geographic risk assessment should consider both how often incidents occur and how severe their consequences are.

Relevance to Nigeria

This approach is particularly useful when assessing a geographically dispersed pipeline network. Applying a similar framework to Nigerian pipeline data could help distinguish between areas with frequent incidents and areas where fewer incidents may still present significant environmental, financial, or safety consequences. For example, a corridor with frequent lower consequence incidents may require stronger monitoring and preventive maintenance, while an area with fewer but potentially high consequence failures may require greater emphasis on engineering assessment, containment, and emergency response.

The key lesson is that geographic risk should be prioritised based on both frequency and consequence, not incident count alone. This would provide a stronger basis for directing

limited integrity-management and emergency-response resources towards the areas where they can have the greatest risk-reduction impact.

Investigation 3: Asset and Infrastructure Risk Assessment

The purpose of this investigation was to understand which types of pipeline assets and infrastructure are most frequently represented in reported incidents. Looking beyond geographic location allowed the analysis to examine whether incident exposure was concentrated around particular pipeline classifications, facility types, commodities, asset ages, or engineering characteristics.

The findings showed that incidents occurred across different parts of the pipeline network, with onshore pump and meter stations, pipeline segments, and terminal facilities accounting for the largest share of reported events. The analysis also showed that incidents occurred across both newer and older assets, while commodity type and other engineering characteristics provided additional context. Importantly, these patterns do not establish that a particular asset characteristic causes failure, especially where the total number of assets within each category is unknown.

Relevance to Nigeria

The key lesson is that pipeline integrity should be managed across the entire infrastructure system rather than focusing only on the pipeline itself. A similar assessment of Nigerian assets could help identify whether pump stations, terminals, storage facilities, pipeline segments, or other supporting infrastructure are receiving sufficient integrity attention. The findings also suggest that characteristics such as asset age, commodity, and pipe diameter are better treated as risk indicators rather than standalone measures of failure risk. Combining these factors with inspection history, operating conditions, maintenance records, location, and previous incidents would provide a more reliable basis for prioritising assets.

This approach would support a more targeted integrity strategy by helping operators identify which assets require closer inspection, stronger monitoring, or additional maintenance based on their overall risk profile rather than relying on a single characteristic.

Investigation 4: Failure Mechanism and Leak Characteristics

The purpose of this investigation was to move beyond understanding where pipeline incidents occur and examine why they occur and what happens when they do. The previous investigations identified the geographic areas and types of infrastructure most frequently represented in the incident records. However, knowing where incidents occur is only part of the integrity picture. Effective pipeline integrity management also requires an understanding of the failure mechanisms behind those incidents and the level of damage they can create.

This investigation therefore examined the major failure mechanisms associated with reported incidents, the characteristics of the resulting leaks, and the financial, environmental, and human safety consequences associated with different types of failure. The objective was not simply to identify the most common causes, but to determine whether the mechanisms occurring most frequently were also the ones creating the greatest overall risk.

◆ **Failure Mechanism Profile:**

The analysis showed that pipeline incidents were concentrated around a relatively small number of recurring failure mechanisms. Equipment failure was the most frequently reported, accounting for 2,702 incidents (45.47%). Corrosion failure followed with 1,299 incidents (21.86%), while incorrect operation accounted for 823 incidents (13.85%). Together, these three mechanisms represented more than 81% of all reported incidents. This is an important finding because it suggests that pipeline reliability may be significantly improved by focusing attention on a relatively small group of recurring problems rather than treating every incident as an isolated event.

Equipment failures point to the importance of maintaining the reliability and condition of pipeline components and associated equipment. Corrosion remains a major long-term integrity concern because deterioration can progressively weaken assets if it is not identified and controlled. Incorrect operation, meanwhile, demonstrates that pipeline integrity is not determined by physical assets alone; operating procedures, workforce competence, and adherence to established controls can also influence system performance. Other mechanisms, including material failure of pipe or weld, natural force damage, excavation damage, and other outside force events, occurred less frequently but became particularly important when their consequences were examined.

◆ **Leak Characteristics:**

The investigation also examined how pipeline failures typically manifest once an incident occurs. Pinhole leaks were the most frequently reported leak type, accounting for 1,397 incidents (23.51%), closely followed by seal or packing failures, with 1,341 incidents (22.57%). Connection failures accounted for another 760 incidents (12.79%), while cracks represented 580 incidents (9.76%). These patterns provide additional insight into the physical nature of pipeline failures. Pinhole leaks and seal or packing failures can be associated with deterioration, corrosion, component wear, or degradation of sealing materials. This reinforces the importance of inspection and condition monitoring capable of identifying deterioration before it develops into a larger failure.

However, approximately 20.63% of the records did not contain a reported leak type. This represents an important data-quality limitation. While the missing information does not prevent the broader incident analysis, it reduces the level of detail available for understanding how failures develop and which physical failure patterns may be recurring.

For a risk analytics system, this is significant because the quality of future predictions and integrity assessments depends partly on the quality of historical information. Improving the completeness and consistency of incident reporting would therefore strengthen the organisation's ability to identify emerging failure patterns over time.

◆ **Financial Consequences:**

The financial analysis produced one of the most important findings from this investigation: The failure mechanism responsible for the greatest number of incidents was not the mechanism responsible for the greatest financial loss.

Material failures involving pipes or welds generated approximately \$2.11 billion in total financial losses, making them the most financially consequential failure mechanism in the dataset. Corrosion followed with approximately \$716.5 million, while equipment failure generated approximately \$422.0 million. This is significant because equipment failure occurred more than three times as frequently as material failure of pipe or weld, yet its total financial impact was substantially lower. The finding demonstrates why incident frequency should not be used as the sole measure of pipeline risk. A failure mechanism may occur relatively infrequently but still create substantial costs through extensive repairs, product loss, environmental remediation, operational disruption, and other associated consequences.

For integrity management, this means that resources should not simply be directed towards the mechanisms with the highest incident counts. High-consequence failure mechanisms also require attention because preventing even a small number of severe events can have a substantial effect on overall business risk.

◆ **Environmental Consequences:**

A similar pattern emerged when environmental impact was examined. Material failure of pipe or weld resulted in the largest volume of unrecovered product, at approximately 191,767 barrels, followed closely by excavation damage, which resulted in approximately 181,932 barrels of unrecovered product. Equipment failure accounted for approximately 125,945 barrels, while corrosion accounted for approximately 74,059 barrels.

The particularly important finding here is excavation damage. It represented only 3.32% of reported incidents, yet it produced one of the largest environmental exposures in the dataset. This demonstrates that a failure mechanism does not need to occur frequently to represent a significant environmental risk. The scale of the release associated with an individual event can be more important than the number of times that mechanism appears in the incident records. Consequently, environmental risk assessment should consider both how often a failure occurs and how much product could potentially be released when it does occur.

◆ **Human Safety Consequences:**

The safety analysis provided another perspective on the difference between frequency and consequence. Excavation damage was associated with the highest number of fatalities, with 8 fatalities, while incorrect operation was associated with the highest number of injuries, with 24 injuries. Other outside force damage was also associated with significant safety consequences. By comparison, equipment failure and corrosion were responsible for a much larger proportion of total incidents but were not associated with the highest human safety consequences in this dataset.

This reinforces an important principle for pipeline risk management: the most common failure is not necessarily the most dangerous failure.

A risk-based integrity strategy therefore needs to consider several dimensions simultaneously. Reducing frequent equipment failures may significantly improve reliability, while controlling excavation damage and operational errors may have a more direct effect on preventing severe human consequences. Similarly, addressing pipe and weld integrity may be particularly important from a financial and environmental perspective.

Overall Interpretation:

Taken together, the findings show that pipeline integrity cannot be managed effectively through a single maintenance strategy. Equipment failure, corrosion, and incorrect operation provide the greatest opportunities for reducing the overall frequency of incidents because they account for the majority of reported failures. However, material failures of pipes and welds, excavation damage, and operational errors demonstrate that lower-frequency mechanisms can still create substantial financial, environmental, or safety consequences. The investigation therefore supports a more comprehensive risk framework in which failure mechanisms are assessed according to both likelihood and consequence.

This distinction is important for Keystra Energy Group because it allows integrity resources to be aligned with the type of risk being addressed. Frequent equipment failures may call for stronger preventive maintenance, condition monitoring, and reliability programmes. Corrosion requires sustained inspection and corrosion-control strategies. Pipe and weld failures may require more advanced integrity assessments and non-destructive testing. Excavation damage requires stronger controls around activities near pipeline infrastructure, while incorrect operation requires attention to procedures, training, competency, supervision, and operational discipline.

The objective is therefore not simply to reduce the number of incidents. It is to reduce the overall risk created by those incidents, with particular attention to failures capable of producing severe consequences.

Relevance to Nigeria

The main lesson that can be applied to Nigeria is the need to understand pipeline risk at the failure mechanism level, rather than relying only on incident counts or geographic location. A similar analysis using Nigerian pipeline incident, maintenance, inspection, and operational data could help identify which failure mechanisms occur most frequently within the local operating environment and which ones have the greatest consequences. This would provide a stronger basis for determining where inspection, maintenance, engineering controls, workforce interventions, and emergency preparedness should be concentrated.

The U.S. findings also demonstrate the importance of separating frequency from severity. A mechanism that occurs frequently may represent the greatest opportunity for improving day-to-day reliability, while a less frequent mechanism may require significant attention because of its potential to cause major environmental, financial, or safety consequences.

For pipeline operations in Nigeria, this approach could be particularly valuable because integrity resources are not unlimited. Rather than applying the same level of intervention across every asset and failure type, operators could use available data to identify the mechanisms that represent the greatest combination of likelihood, consequence, and exposure. The lesson is therefore not that the same failure mechanisms identified in the U.S. dataset necessarily represent the same level of risk in Nigeria. Instead, the value lies in applying the analytical framework: identify the failure mechanisms, understand how frequently they occur, assess the consequences they create, and use that information to determine the most appropriate integrity controls.

Local Nigerian data would ultimately be required to validate which failure mechanisms are most significant in the Nigerian pipeline environment. However, the U.S. analysis demonstrates how historical incident data can be transformed from a record of past failures into a practical foundation for more proactive, risk-based pipeline integrity management.

Investigation 5: Consequence Assessment and Business Impact

The purpose of this investigation was to assess the financial, environmental, human, and overall business consequences of pipeline incidents, moving beyond the causes of failure examined in the previous investigation.

The analysis estimated approximately \$4.31 billion in total incident-related costs, with emergency response (\$1.77 billion) and environmental remediation (\$1.67 billion) accounting for the largest shares. Property damage contributed \$672.12 million, while product loss accounted for \$61.33 million. This shows that the cost of a pipeline incident extends far beyond the value of the product lost. Approximately 1.34 million barrels of hazardous liquid were released, with about 543,929 barrels recovered and approximately 799,003 barrels remaining unrecovered. The dataset also recorded 16 fatalities and 46

injuries. Although most incidents did not result in casualties, the findings reinforce the importance of protecting workers and the public alongside managing financial and environmental risks.

More than 70% of incidents were classified as low impact, while 22.82% were medium impact and 6.73% were high impact. Although high-impact incidents were less frequent, they can create disproportionately large financial, environmental, and operational consequences. This demonstrates that pipeline risk cannot be measured by incident frequency alone; the severity and potential consequences of failure must also guide integrity priorities.

Relevance to Nigeria

The key lesson for Nigeria is the importance of assessing pipeline risk based on both frequency and consequence. Applying a similar framework to local data could help identify incidents and operating conditions with the greatest financial, environmental, operational, and safety impacts. The findings also highlight that integrity management should extend beyond prevention. Early leak detection, effective containment, emergency response, and recovery can significantly reduce the consequences when a failure occurs.

While Nigerian data would be needed to establish the local risk profile, the U.S. analysis provides a useful framework for prioritising resources around both preventing failures and reducing their consequences.

Investigation 6: Strategic Risk Intelligence and Decision Framework

The purpose of this final investigation was to consolidate the findings from the previous analyses and translate them into practical pipeline integrity priorities. The earlier investigations established where incidents occur, which assets are most exposed, what failure mechanisms are involved, and the consequences they create. This investigation brings those insights together to determine where integrity efforts should be focused.

The combined findings show that pipeline risk is not evenly distributed. Certain geographic areas and infrastructure types recorded higher incident exposure, while equipment failure and corrosion were the leading contributors to incident frequency. At the same time, pipe and weld failures, excavation damage, and operational errors demonstrated that less frequent failures can still create significant financial, environmental, and safety consequences. These findings support a shift from routine maintenance towards a risk-based integrity strategy, where inspection and maintenance priorities are determined by a combination of likelihood, asset criticality, and potential consequence.

Strategic Priorities:

Five priorities emerged from the analysis:

- 1. Risk-Based Geographic Inspection —** Prioritise inspections in areas with higher incident and consequence exposure.
- 2. Critical Asset Reliability —** Focus integrity efforts on key pipelines, pump stations, and terminal facilities.
- 3. Preventive Maintenance —** Strengthen equipment reliability and corrosion management to address recurring failures.
- 4. High Consequence Risk Management —** Apply targeted engineering and operational controls to failures capable of causing severe impacts.
- 5. Emergency Preparedness —** Strengthen leak detection, containment, and emergency response to minimise consequences when failures occur.

Relevance to Nigeria

The key lesson for Nigeria is the importance of turning pipeline data into actionable risk priorities. A similar framework could combine local incident history, asset condition, failure mechanisms, operational exposure, and consequence severity to determine where inspection, maintenance, engineering controls, and emergency preparedness should receive greater attention.

PIPELINE RISK DASHBOARDS

Two dashboards were developed to translate the project's analytical findings into practical pipeline risk insights.

- 1. Executive Pipeline Risk Overview —** designed for senior management to monitor overall pipeline performance, including incident trends, financial exposure, environmental impact, operator performance, and workforce safety.
- 2. Pipeline Integrity and Operational Intelligence —** designed for integrity engineers and maintenance planners to examine asset characteristics, infrastructure components, geographic patterns, failure mechanisms, and product recovery, supporting more targeted inspection and maintenance planning.

Together, the dashboards provide both an executive overview and a deeper operational view of pipeline integrity performance.

PROJECT LIMITATIONS

This analysis has several limitations. First, the findings are based on public U.S. pipeline incident data, so they should not be assumed to represent the Nigerian pipeline environment. Local Nigerian data would be required to validate whether the observed patterns apply in Nigeria. Second, the dataset is primarily incident based and does not provide complete information on the total pipeline population, asset exposure, operating

hours, or inspection history. Therefore, incident counts should not be interpreted directly as failure probabilities.

Finally, some records contain missing or incomplete information, particularly for certain incident and leak characteristics. The solution is therefore presented as a portfolio proof-of-concept, demonstrating an analytical approach rather than a fully deployed predictive or operational system.

PROJECT ARTEFACTS & FULL ACCESS

Available Artefacts:

Full Project Report | Executive Summary | Power BI Report | Dashboard Pictures | SQL Analysis

These are all available through the link below

Github:

https://github.com/winnerEM/Keystra-Energy-Solution-2_Pipeline-Integrity-and-Operational-Risk-Intelligence

Also accessible on

Google drive:

<https://drive.google.com/drive/folders/1ww4sblhMP1151PJ-ZwQHoJ3kmtnXk2F5?usp=sharing>